TWINKLE GROVER

Data Scientist · Predictive Modelling · Fraud & Healthcare Analytics · ML Engineer

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PROFESSIONAL SUMMARY

Data Scientist with 8+ years of cross-domain experience spanning financial services (KYC/AML, credit card analytics) and healthcare claims — demonstrating the ability to extract signal from fundamentally different data environments within the same organisation. Built and deployed four end-to-end ML systems: XGBoost healthcare cost prediction on 100K+ CMS Medicare records, Random Forest insurance fraud detection with Flask REST API, real-time n8n fraud automation pipeline, and a SQL Server Data Warehouse using Medallion Architecture. Proficient in Python (XGBoost, Scikit-learn, Pandas), SQL, and data visualisation (Tableau, Power BI). Proven ability to adapt quickly to new problem spaces, collaborate cross-functionally, and deliver production-quality research — suited for interdisciplinary roles requiring both quantitative rigour and domain agility.

TECHNICAL SKILLS

Machine Learning & AI: XGBoost, Random Forest, Logistic Regression, Scikit-learn, Feature Engineering, EDA, Threshold Optimisation, Class Balancing, Predictive Modelling

Python Stack: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Flask API, Jupyter Notebook

Data Engineering: SQL Server, PostgreSQL, SAS, ETL Pipelines, Medallion Architecture (Bronze/Silver/Gold), Star Schema, Data Warehouse Design, SSMS

Automation & BI: n8n (Webhook/API/Conditional Routing), Tableau, Power BI, Excel (VBA/Macros), Google Sheets

Quantitative / Domain: Financial Analytics, Alternative Data, Healthcare Claims Analytics, Insurance Fraud Detection, KYC/AML Compliance, Credit Card Analytics, Revenue Assurance, Econometric Modelling

Collaboration: Cross-time-zone remote teamwork, stakeholder documentation, research communication, data storytelling

PROFESSIONAL EXPERIENCE

Senior Business Analyst | EXL — *May 2017 – Present*

↳ Healthcare Analytics — *2021 – Present*

Transitioned into healthcare data analytics, applying ML and SQL expertise to claims data, cost modelling, and operational reporting — demonstrating rapid domain adaptation from financial services to a structurally different regulated industry.

- Designed SQL pipelines for healthcare claims reporting and cost analytics, handling large-scale inpatient/outpatient datasets across multiple coverage dimensions.
- Applied EDA and feature engineering techniques to healthcare datasets — foundation for the CMS Medicare XGBoost cost prediction project (100K+ records, multi-dimensional cost aggregation).
- Delivered Tableau dashboards surfacing claims trends, error patterns, and reimbursement anomalies for operational and compliance stakeholders.

↳ Banking, KYC/AML & Credit Card Analytics — *2017 – 2021*

Core domain: financial data engineering, compliance automation, and credit card analytics — extracting signal from transactional, regulatory, and behavioural datasets in a high-stakes banking environment.

- Engineered a SQL ETL validation engine with 4 stored procedures and 40+ automated KYC/CRR rules, including audit trail, error logging, and automated CRRF master table loading.
- Visualised KYC error patterns and data quality trends in Tableau, enabling real-time compliance team action on flagged anomalies.

- Designed SQL pipelines for Citi Cards customer segmentation, profiling, and monthly performance reporting; performed large-scale EDA to detect revenue leakage, billing errors, and misaligned transactions.
- Conducted COGS, revenue, and cost-driver analysis against P&L statements (IRMS/CVS Health); designed macro-automated Excel dashboards with CSV ingestion (Marpai).
- Built VBA employee database automation tool with validation and staffing integration; managed 50+ state-level regulatory filings and SLA tracking across insurance domain projects.

DATA SCIENCE & RESEARCH PROJECTS

Healthcare Cost Prediction — CMS Medicare Claims — *Python · XGBoost · Pandas · Scikit-learn · Matplotlib*

github.com/kristlingr/CMS-SynPUF-US-Healthcare-Claims-Fraud

- Ingested and processed 100,000+ CMS DE-SynPUF Medicare records, merging beneficiary and carrier claim lines across inpatient, outpatient, and carrier cost dimensions — representative of large-scale alternative data analysis.
- Engineered domain features including TOTAL_CHRONIC (11 chronic conditions), cost aggregates, FULL_YEAR_COVERAGE flag, and beneficiary age; applied log1p transformation on skewed reimbursement targets.
- Trained XGBoost model (lr=0.05, max_depth=6, 500 rounds with early stopping) outperforming Linear Regression baseline on MAE, RMSE, and R^2; generated feature importance and Actual vs. Predicted visualisations.

End-to-End Insurance Fraud Detection System — *Python · Scikit-learn · Flask · n8n · Tableau*

github.com/kristlingr/Insurance-Fraud-detection-Model-System

- Built complete ML pipeline: EDA, feature engineering (claim deviation, claim ratio, policy age), balanced Random Forest with custom threshold (0.2) to maximise fraud recall; segmented claims into High/Medium/Low risk tiers.
- Deployed model as Flask REST API; orchestrated claim intake via n8n; delivered Tableau dashboards with ROC-AUC curve, feature importance, and fraud trend reporting.

Real-Time Fraud Detection Pipeline — n8n Automation — *n8n · Python · Outlook · Google Sheets*

github.com/kristlingr/insurance-fraud-detection-n8n

- Designed two-version automated fraud pipeline: v1 (rule-based Python logic) upgraded to v2 (ML API-integrated real-time scoring); webhook ingestion with conditional Outlook alerts and full Google Sheets audit log.
- Demonstrates ability to write production-quality, live-system code integrating ML research into automated workflows.

SQL Server Data Warehouse — Medallion Architecture — *SQL Server · SSMS · Star Schema · ETL*

github.com/kristlingr/SQL-DataWarehouse_project

- Architected modern data warehouse: Bronze (raw CSV ingestion), Silver (cleansing, standardisation), Gold (star schema with fact/dimension tables optimised for analytics).
- Built ETL pipelines and SQL analytics reporting on customer behaviour, product performance, and sales trends; documented data model for business and analytics stakeholders.

EDUCATION

Bachelor of Commerce (B.Com) | University of Delhi *2016*

KEY ACHIEVEMENTS

- Designed and shipped 4 production ML systems independently — from raw data ingestion to deployed API and live dashboards.
- Processed 100K+ Medicare records to build a predictive healthcare cost model, applying domain-aware feature engineering.
- Automated real-time fraud detection pipeline using n8n and Flask, demonstrating ability to integrate research into live systems.
- Delivered 40+ KYC/AML automated compliance rules with full audit trail, reducing manual validation overhead.

- Managed 50+ state-level regulatory filings end-to-end, demonstrating operational rigour alongside quantitative output.